MIT | Academy of Engineering

(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

THEORY OF COMPUTATION (Course Code: 2304220T)
GROUP MINI-PROJECT ACTIVITY
REGEX-Based Web Form Validation

REPORT ON

Resume Parser for Campus Placement

MIT Academy of Engineering

(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

Alandi (D), Pune– 412105 (2025–2026)

Student Details

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Name of Faculty/Guide : Dr.Rutuja Kadam

Link: https://om2o4.github.io/TOC-Regex/

Topic: Resume Parser for Campus Placement (Regex-Based Validation System)

Project Description

This project focuses on designing and implementing a Resume Parser and Validation System using Regular Expressions (Regex) as part of the Theory of Computation course. The system simulates a real-world campus placement portal where candidate details are validated and structured data is extracted from resume input.
The web-based application allows users to enter personal, academic, and professional details such as Name, Email, Phone Number, Skills, CGPA, Graduation Year, and LinkedIn profile. Additionally, users can paste resume text, from which the system automatically extracts key information like email and phone number using pattern matching.
The project demonstrates how formal language theory concepts (regular languages, DFA/NFA) are applied in practical software systems like recruitment platforms and Applicant Tracking Systems (ATS).

Objectives

- To design a web-based validation system using regex
- To apply formal language concepts (regular expressions and automata)
- To ensure accurate validation and error handling
- To simulate a real-world resume processing system

Technologies Used

- HTML5– Form structure
- CSS3– UI design and styling
- JavaScript (ES6)– Validation logic and data extraction
- Regex (Regular Expressions)– Pattern matching and validation

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System Features

- Validation of 7+ input fields using regex
- Real-time user feedback with error messages
- Resume text parsing for extracting email and phone number
- Clean and user-friendly interface
- Edge-case handling for invalid inputs

Team Members & Contribution

Sr. No	Name	Role	Work Description
1	Om Mule	Team Lead & Backend (Validation)	Designed regex patterns, implemented validation logic, handled edge cases,Report
2	Vikrant Haral	UI Developer	Designed layout, styling, responsive UI, improved user experience,Test
3	Shreyas Salunkhe	Script Developer	Implemented JavaScript logic, real-time validation, data extraction,DA Diagram

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Challenges Faced

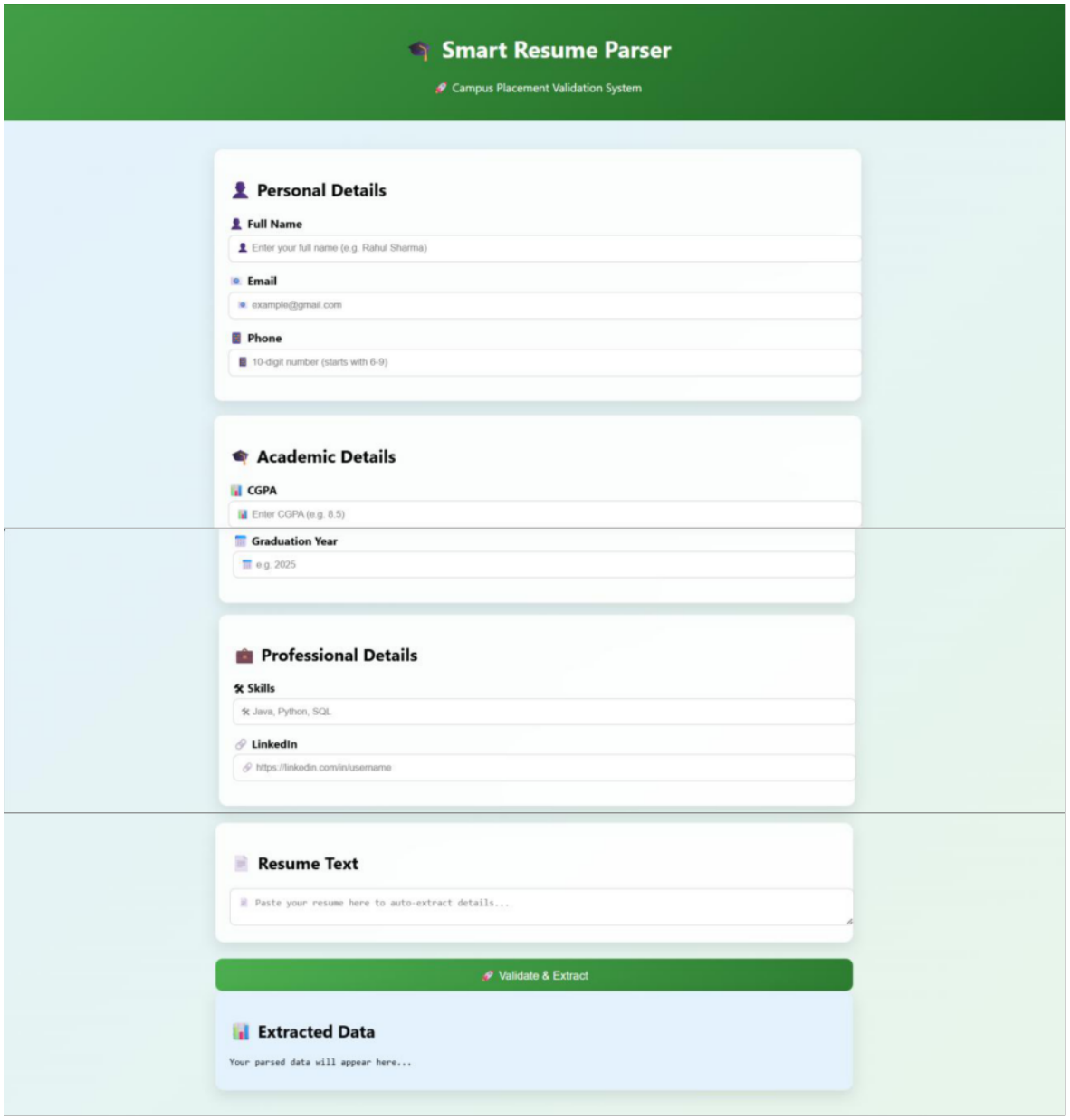
- Designing accurate regex patterns for different input formats
- Handling edge cases such as invalid characters and partial inputs
- Ensuring real-time validation without performance issues
- Understanding the mapping between regex and automata (DFA/NFA)

Key Learnings

- Practical application of regular expressions in real systems
- Understanding how finite automata power validation engines
- Importance of efficient pattern design to avoid redundancy
- Improved skills in frontend development and debugging
- Exposure to real-world software design concepts

Conclusion

The project successfully demonstrates how theoretical concepts from the Theory of Computation can be applied to real-world problems like resume parsing and data validation. The system ensures correctness, efficiency, and usability, making it a practical example of regex-based validation in modern web applications.


Smart Resume Parser
Campus Placement Validation System
Personal Details
Full Name
Enter your full name (e.g. Rahul Sharma)
Email
example@gmail.com
Phone
10-digit number (starts with 6-9)
Academic Details
CGPA
Enter CGPA (e.g. 8.5)
Graduation Year
e.g. 2025
Professional Details
Skills
Java, Python, SQL
LinkedIn
https://linkedin.com/in/username
Resume Text
Paste your resume here to auto-extract details...
Validate & Extract
Extracted Data
Your parsed data will appear here...